Inverse Problems in Geophysics Part 3: Errors and under-determined problems

2. MGPY+MGIN

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Recap

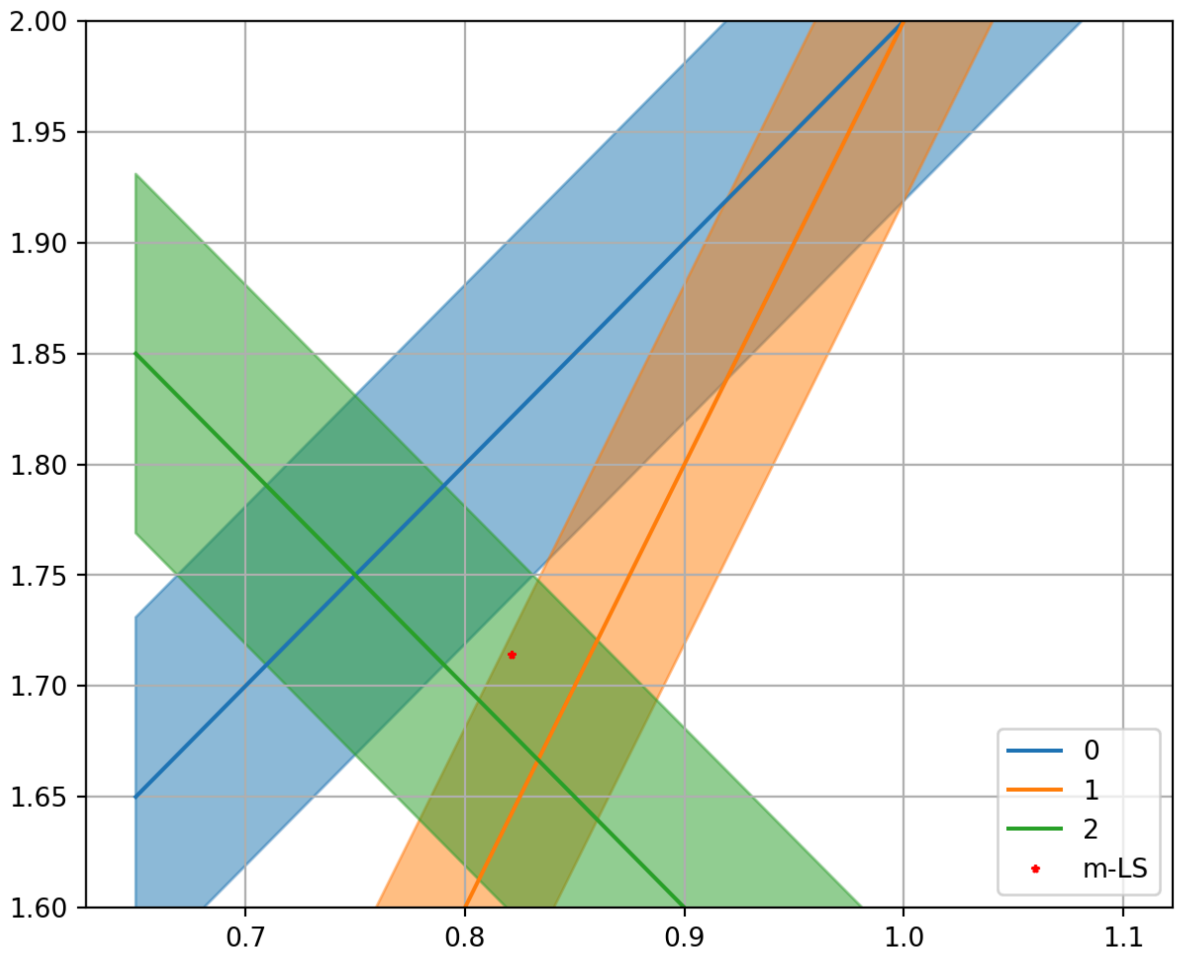
- minimization of objective function (L2 norm of error-weighted misfit)
- least-squares solution for over/even-determined problems
- least-squares inverse operator $\mathbf{m}_{LS} = \mathbf{G}^{\dagger}\mathbf{d}$
  with $\mathbf{G}^{\dagger} = (\mathbf{G}^T\mathbf{G})^{-1}\mathbf{G}^T$ ($M \times M$ x $M \times N$=$M \times N$)
- error weighting by multiplying $\mathbf{G}$ and $\mathbf{d}$ with $\mathrm{diag}(1/\mathbf{e})$
- resolution matrices for model (perfect) and data (distributed)

Simple matrix problem

Simple matrix problem - Solution

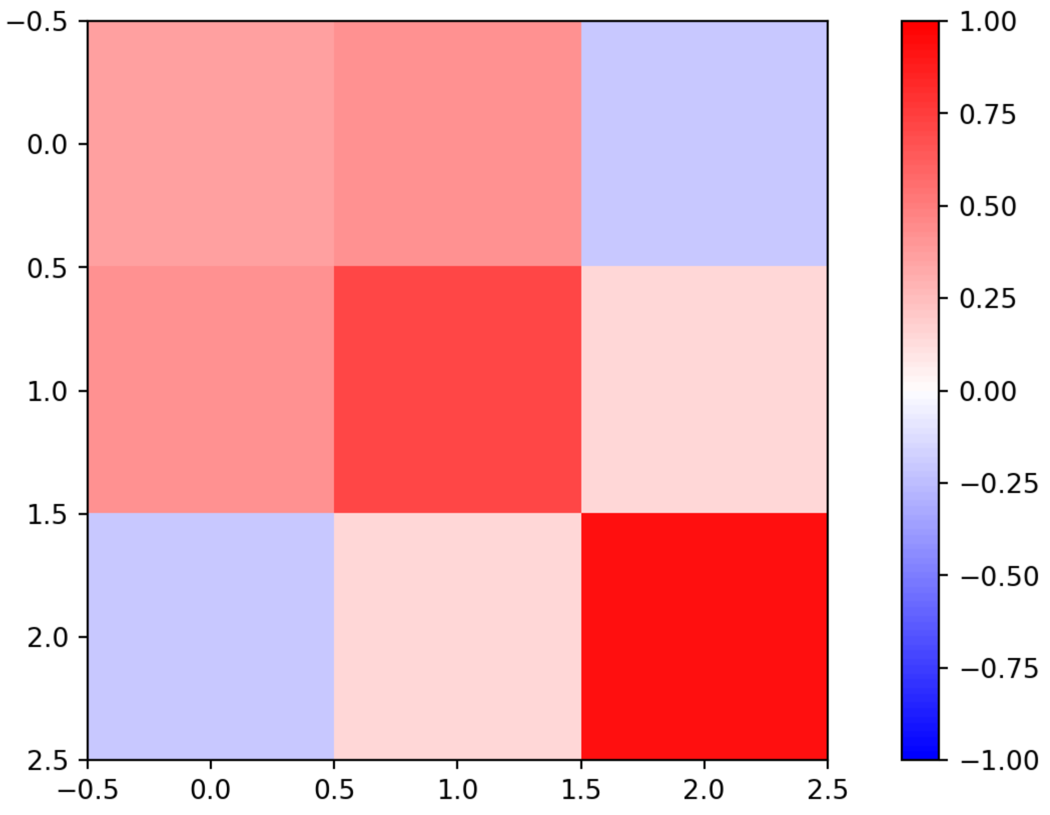
```
mLS, *_ = np.linalg.lstsq(G, d, rcond=None)
fig, ax = plt.subplots(figsize=(10, 6))
x = np.array([0.65, 1.1])
dd = 0.081
for i in range(len(G)):
    y = (d[i] - G[i, 0] * x) / G[i, 1]
    dy = dd/G[i, 1]
    co = f"C{i}"
    ax.fill_between(x, y-dy, y+dy,
        color=co, alpha=.5)
    ax.plot(x, y, "-", label=f"{i}", color=co)

ax.plot(mLS[0], mLS[1], "r*", label="m-LS")
ax.set_aspect(1.0)
ax.set_ylim(1.6, 2.0)
ax.legend()
ax.grid()
```



Simple matrix problem - Resolution

```
Ginv = np.linalg.inv(G.T @ G) @ G.T
RM = Ginv@G
RD = G@Ginv
im = plt.imshow(RD, vmin=-1, vmax=1, cmap="bwr")
plt.colorbar(im)
```

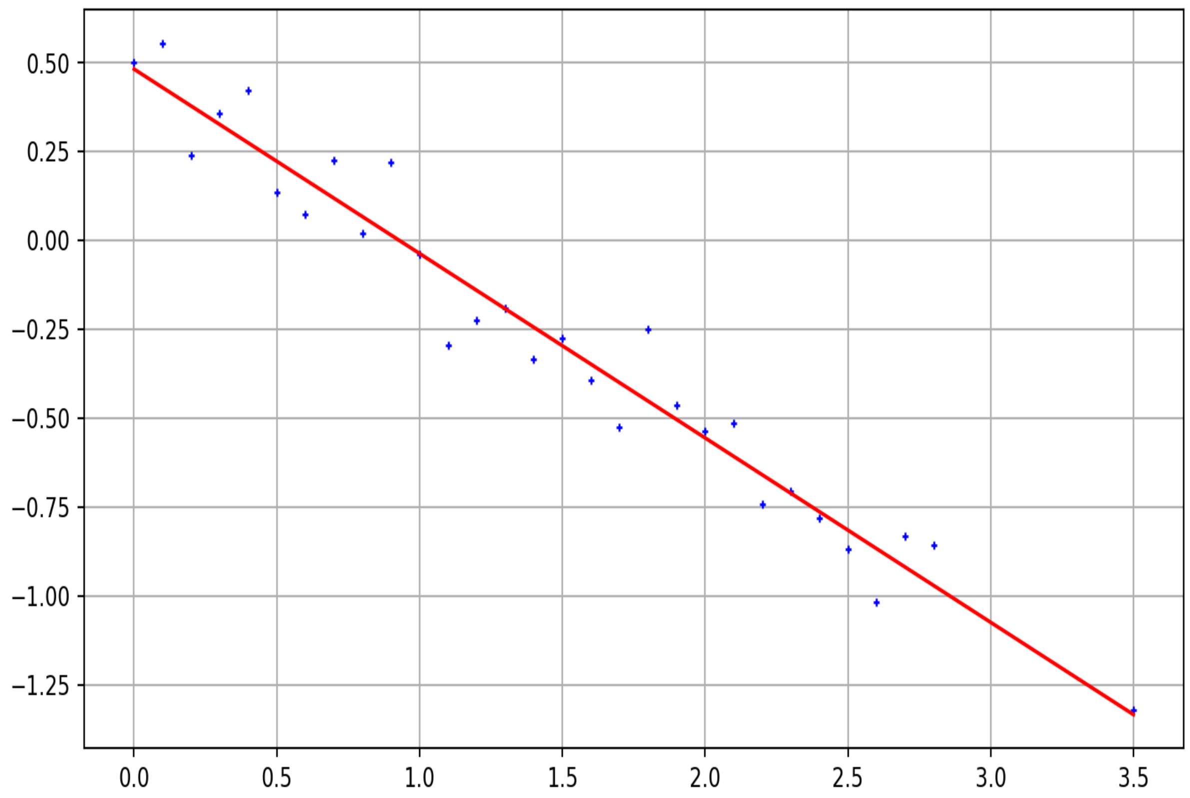


- 3rd row (measurement) is most important
- 1st and 2nd are highly correlated, `sum(diag(RD))` $\Rightarrow$ 2

Linear regression

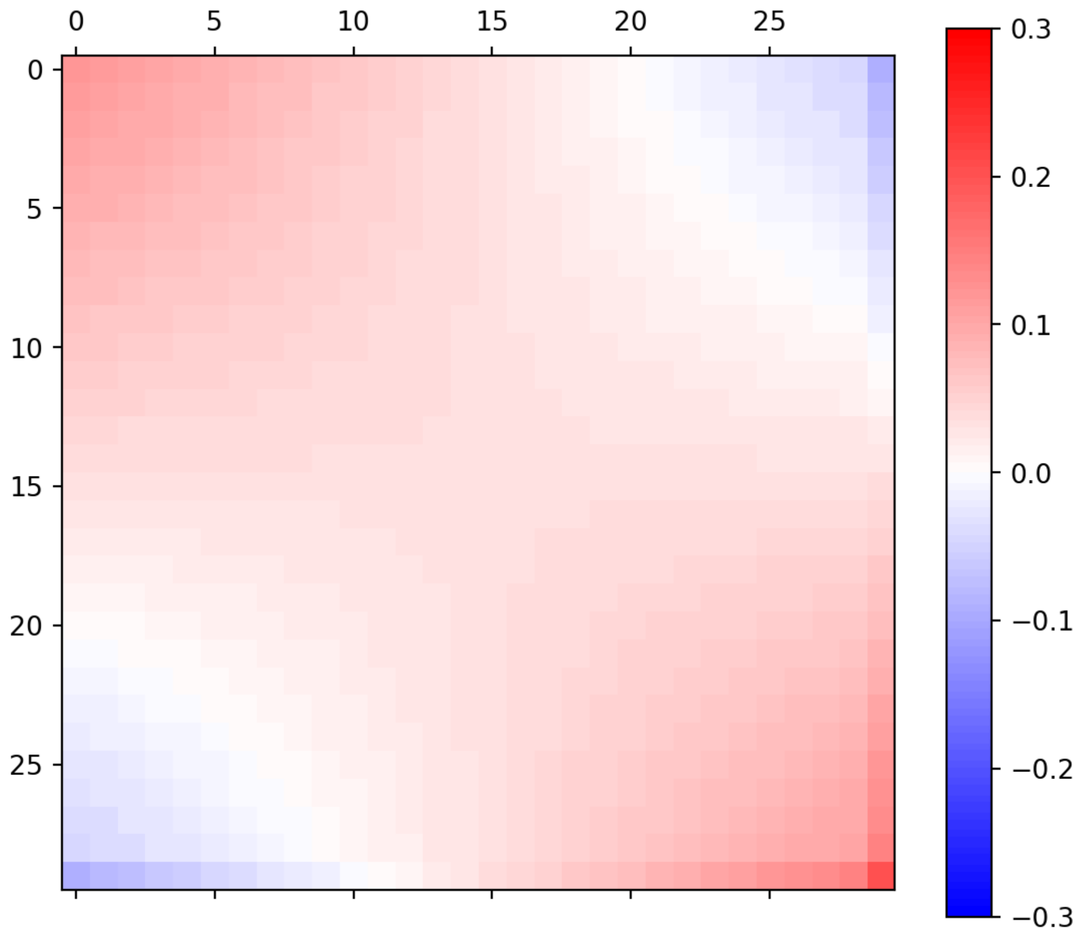
```
x = np.arange(0, 3, 0.1)
x[-1] = 3.5
err = 0.1
d = 0.5-0.5*x + np.random.randn(len(x)) * err
A = np.ones((len(x), 2))
A[:, 1] = x[:]
Ainv = np.linalg.inv(A.T @ A) @ A.T
m = Ainv @ d
print("m = ", m)
fig, ax = plt.subplots()
ax.plot(x, d, "b+")
ax.plot(x, A@m, "r-")
ax.grid()
residual = A @ m - d
print("mean=", np.mean(residual))
print("std=", np.std(residual))
print("X²=", np.sqrt(np.mean((residual/err)**2)))
```

```
m =  [ 0.48226107 -0.51893123]
mean= 4.996003610813204e-17
std= 0.10178870098811599
X²= 1.0178870098811599
```



Linear regression - data importance matrix

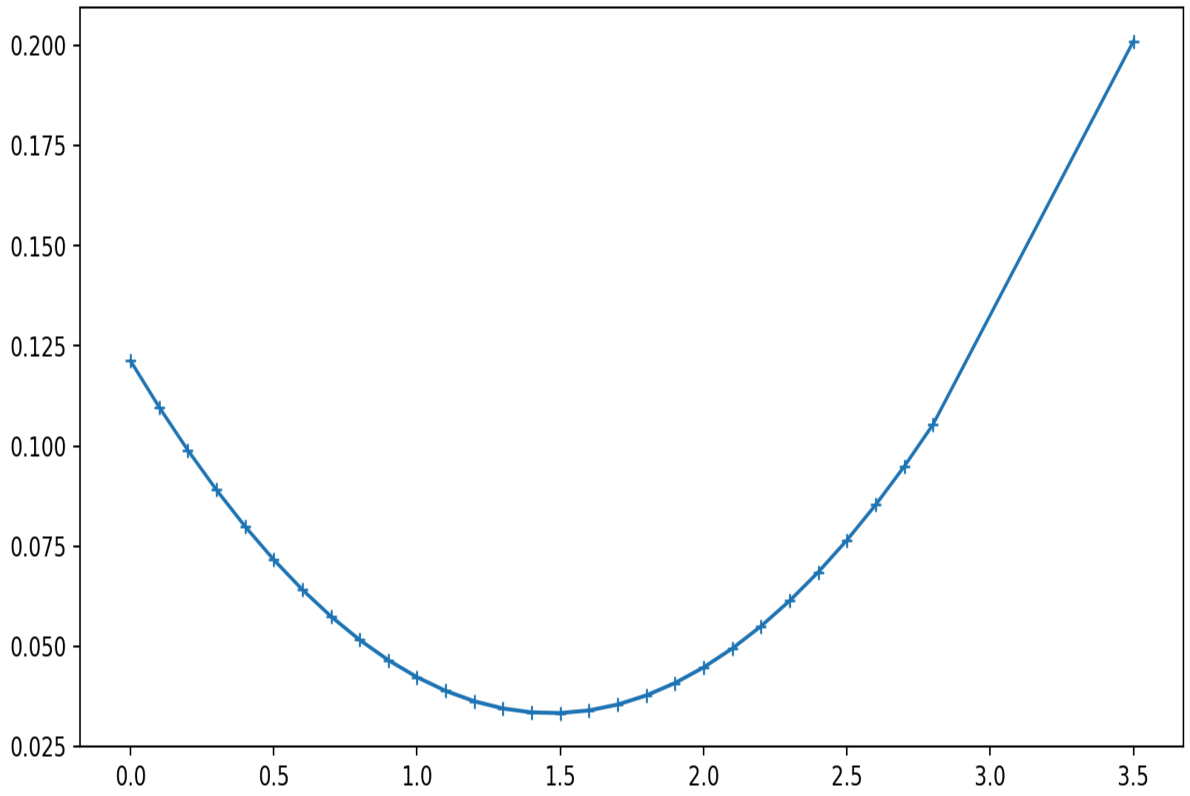
```
RD = A @ Ainv
fig, ax = plt.subplots(figsize=(7,6))
im = ax.matshow(RD, cmap="bwr",
    vmin=-0.3, vmax=0.3)
plt.colorbar(im);
```



- neighboring data points are correlated, opposite anti-correlated

```python
rd = np.diag(RD)
display(np.sum(rd))
plt.plot(x, rd, "+-", ms=5);
```

1.9999999999999993



- main diagonal of $\mathbf{R}^D \Rightarrow$ **data importance** (outside biggest)
- sum of importances equals M (later: rank of matrix)

Errors & robust inversion

- error goes into the scaling of $\mathbf{G}$ and $\mathbf{d}$
- sometimes data have different errors
- there might be outliers spoiling the result

Error weighting

Unweighted residual norm (root-mean square RMS)

$$\|\mathbf{d} - \mathbf{f}(\mathbf{m})\| = \sqrt{1/N \sum (d_i - f_i(\mathbf{m}))^2}$$

Weighting by individual error ϵ_i (chi-square value):

$$\chi^2 = \frac{1}{N} \sum \left(\frac{d_i - f_i(\mathbf{m})}{\epsilon_i} \right)^2 \to \min$$

In case of exact error estimates: $\chi^2 = 1$

Error weighting

replace d_i by $\hat{d}_i = d_i/\epsilon_i$ leads to

$$\mathbf{m} = (\hat{\mathbf{G}}^T\hat{\mathbf{G}})^{-1}\hat{\mathbf{G}}^T\hat{\mathbf{d}}$$

with $\hat{\mathbf{G}} = \mathrm{diag}(1/\epsilon_i) \cdot \mathbf{G}$

General L_p norm

The L_2 norm is the rooted sum of squared elements of a vector $\mathbf{v}$:

$$\Phi^P = \|\mathbf{v}\|_2 = \sqrt{\sum_i v_i^2}$$

More generally, the L_P norm is computed by

$$\Phi^P = \|\mathbf{v}\|_P = \left(\sum_i |v_i|^P\right)^{1/P}$$

L_1 norm minimization (IRLS)

$$\Phi^1 = \|\mathbf{v}\|_1 = \sum_i |v_i|$$

The L_1 norm is much less prone to outliers as the misfit is not squared

By the ratio of its contribution to L1 and L2 norm we can reweight ϵ_i

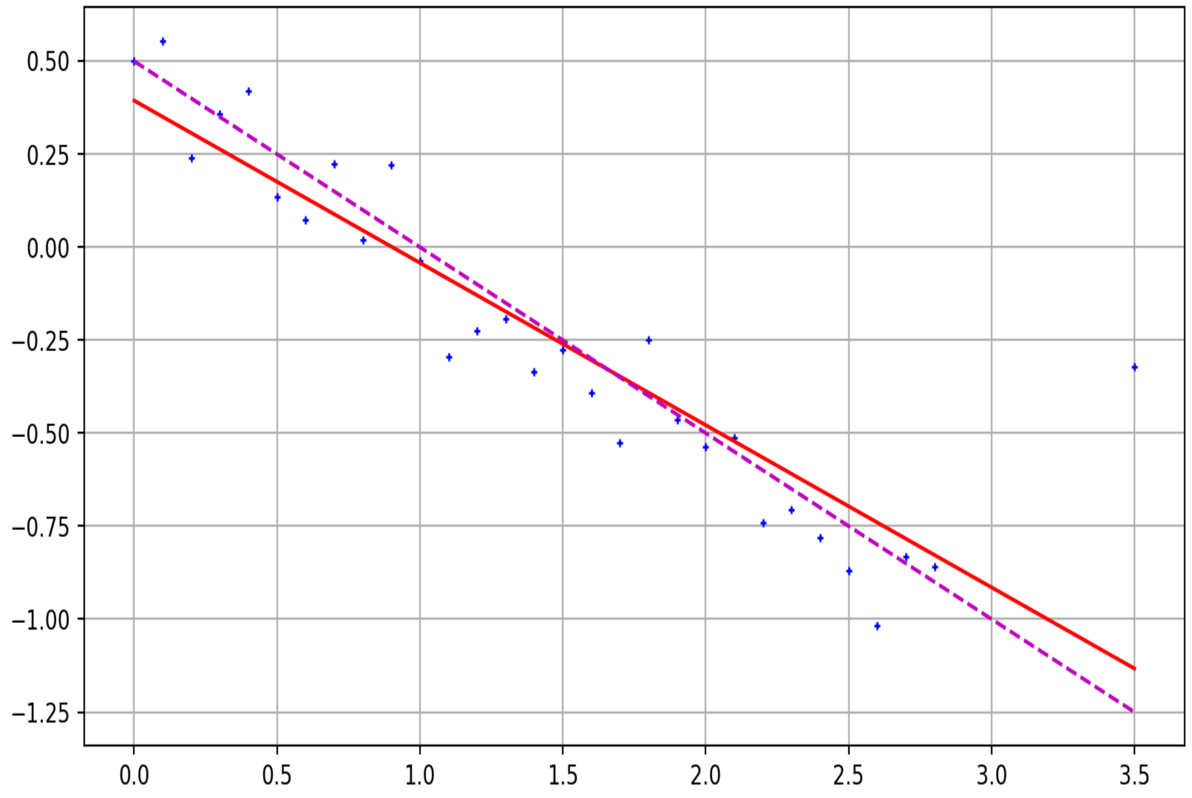
$$w_i = \frac{|v_i| / \|\mathbf{v}\|_1^1}{v_i^2 / \|\mathbf{v}\|_2^2} = \frac{\|\mathbf{v}\|_2^2}{|v_i| \cdot \|\mathbf{v}\|_1^1}$$

This is known as **Iteratively Reweighted Least-Squares** (IRLS)

Linear regression with outliers

```
d[-1] += 1
m = Ainv @ d
print("m = ", m)
fig, ax = plt.subplots()
ax.plot(x, d, "b+")
ax.plot(x, A@m, "r-")
ax.plot(x, 0.5-0.5*x, "m--")
ax.grid()
```

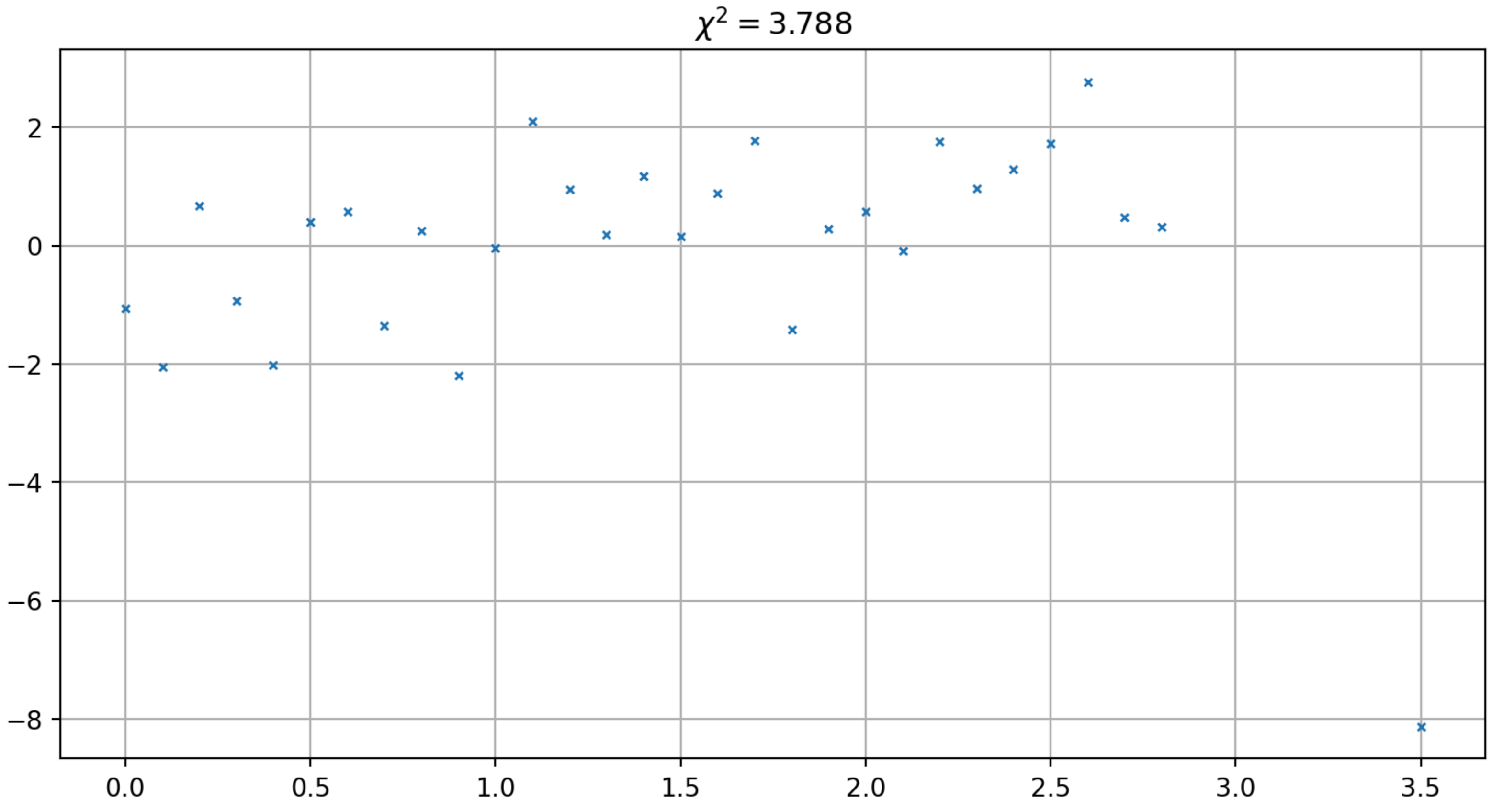
```
m =  [ 0.3941068 -0.4362866]
```



The misfit distribution

Important

Have a look at the misfit distribution (not only the value)



Under-determined problems

- specifically $N < M$ (too less data for the model pameters), but, more general, if the rank $r < M$

> **Rank of a matrix**
>
> number of independent columns and rows, i.e. the degree of non-degenerateness.

Example of a mixed-determined problem

$$\mathbf{G} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- N<M: clearly under-determined,
- parameter m_3 is perfectly determined (=d_2)
- any combination of m_1 and $m_2 = d_1 - m_1$ fulfils the first equation there is no unique solution (model ambiguity)

Changing the system

$$\mathbf{G} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

adding data 1+2

$$\mathbf{G} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

does not help, because they are linear dependent ($rk(\mathbf{G}) = 2$).

Minimum-norm solution

Analog to the least-squares inverse $(\mathbf{G}^T\mathbf{G})^{-1}\mathbf{G}^T$ there is an inverse for under-determined problems:

$$\mathbf{m} = \mathbf{G}^{\mathrm{T}}(\mathbf{G}\mathbf{G}^{\mathrm{T}})^{-1}\mathbf{d} = \mathbf{G}^{\dagger}\mathbf{d}$$

that is called minimum-norm solution.

Observation from Notebook

The backslash operator (`G\d`) yields the minimum-norm solution.

Derivation

From the models fulfilling $\mathbf{Gm} = \mathbf{d}$ we are looking for the “smallest”

$$\min \Phi = \mathbf{m}^{\mathbf{T}}\mathbf{m} \quad \text{with} \quad \mathbf{Gm} = \mathbf{d}$$

We use the method of Lagrangian parameters (λ_i)

$$\Phi = \mathbf{m}^{\mathbf{T}}\mathbf{m} + \lambda^{\mathbf{T}}(\mathbf{Gm} - \mathbf{d}) \rightarrow \min$$

Derivation (2)

The derivations vanish

$$\frac{\partial \Phi}{\partial \lambda} = \mathbf{Gm} - \mathbf{d} = \mathbf{0}$$

$$\frac{\partial \Phi}{\partial m} = 2\mathbf{m}^{\mathbf{T}} + \lambda^{\mathbf{T}}\mathbf{G} = \mathbf{0}$$

$$\Rightarrow \mathbf{m} = -\frac{1}{2}(\lambda^{T}\mathbf{G})^{T} = -\frac{1}{2}\mathbf{G}^{T}\lambda$$

Derivation (3)

$$\Rightarrow \mathbf{d} = \mathbf{Gm} = \mathbf{G}(-\frac{1}{2}\mathbf{G^T}\lambda) = -\frac{1}{2}\mathbf{GG^T}\lambda$$

from which we can determine the Langrangian parameters

$$\lambda = -2(\mathbf{GG^T})^{-1}\mathbf{d}$$

Replacing them we obtain the minimum-norm solution

$$\mathbf{m}_{MN} = \mathbf{G^T}(\mathbf{GG^T})^{-1}\mathbf{d}$$

Resolution of the minimum norm inverse

$$\mathbf{G}^{\dagger} = \mathbf{G}^{\mathrm{T}}(\mathbf{G}\mathbf{G}^{\mathrm{T}})^{-1}$$

$$\mathbf{R}^{M} = \mathbf{G}^{\dagger}\mathbf{G} = \mathbf{G}^{\mathrm{T}}(\mathbf{G}\mathbf{G}^{\mathrm{T}})^{-1}\mathbf{G}$$

$$\mathbf{R}^{D} = \mathbf{G}\mathbf{G}^{\dagger} = \mathbf{G}\mathbf{G}^{\mathrm{T}}(\mathbf{G}\mathbf{G}^{\mathrm{T}})^{-1} = \mathbf{I}$$

> **Note**
>
> For underdetermined problems, all data are independent and equally important. Model parameters are insufficiently resolved.

Singular value decomposition

Any matrix $\mathbf{A}$ can be decomposed into model and data eigenvectors, weighted by singular values:

$$\mathbf{A} = \sum_{i=1} r\sigma_i \mathbf{u}_i \cdot \mathbf{v}_i^T = \mathbf{U\Sigma V}^T$$

with the eigenvalues in $\Sigma = \mathrm{diag}(\sigma_i)$, a set of orthogonal data eigenvectors $\mathbf{U} \in R^{N\times N}$ with $\mathbf{U}^{-1} = \mathbf{U}^T$, and model eigenvectors $\mathbf{V} \in \mathbb{R}^{M\times M}$ with $\mathbf{V}^{-1} = \mathbf{V}^T$.

Eigenvectors and singular values

The eigenvectors associated with zero singular values ($\sigma_i = 0$) span the null spaces of data and model. The number of non-zero eigenvalues p defines the rank of the matrix which can be abbreviated by

$$\mathbf{A}_p = \sum_{i=1}^{p} \sigma_i \mathbf{u} \cdot \mathbf{v}^T = \mathbf{U}_p \mathbf{\Sigma}_p \mathbf{V}_p^T$$

where $\mathbf{B}_p$ holds the first p columns of the matrix $\mathbf{B}$.

A matrix can be approximated by choosing the rank by hand.

Forward operator in terms of SVD

The forward problem $\mathbf{Gm} = \mathbf{d}$ can be written in terms of SVD

$$\mathbf{U}_p \boldsymbol{\Sigma}_p \mathbf{V}_p^T \mathbf{m} = \mathbf{d}$$

and rearranged by using $\mathbf{U_p^T U_p} = \mathbf{V_p^T V_p} = \mathbf{I}$ to

$$\boldsymbol{\Sigma}_p \mathbf{V}_p^T \mathbf{m} = \mathbf{U}_p^T \mathbf{d}$$

$$\mathbf{V}_p^T \mathbf{m} = \boldsymbol{\Sigma}_p^{-1} \mathbf{U}_p^T \mathbf{d}$$

Inverse operator in terms of SVD

The generalized inverse $\mathbf{V}_p\mathbf{\Sigma}_p^{-1}\mathbf{U}_p^T$ is called the pseudo-inverse and equals

$$(\mathbf{G}^T\mathbf{G})^{-1}\mathbf{G}^T$$

i.e., the least-squares inverse for overdetermined problems.

By further truncating the number of nonzero singular values one can obtain a regularized solution for ill-posed (badly conditioned under-determined or mixed-determined) problems.

Derivation

The quadratic matrix $\mathbf{A}$ projects a vector in another direction. Special vectors are eigenvectors who keep their direction

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$$

The solution of the equation $\mathbf{A} - \lambda\mathbf{I} = 0$ leads to eigenvalues λ over the characteristic polynome $\det(\mathbf{A} - \lambda\mathbf{I})$ that correspond to eigenvectors in the matrix $\mathbf{Q}$ by

$$\mathbf{A} = \mathbf{Q}\mathrm{diag}(\lambda_i)\mathbf{Q}^T = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T$$